

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	MANOJ KUMAR K	Reg. No.:	192521041
Section / Batch:	2025	Date:	16/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. A virtual reality system renders scenes using the recurrence $T(n) = 6T(n/3) + n \log n$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how rendering performance scales with increasing scene complexity. Interpret efficiency using asymptotic notation.

Q2. Medical Diagnosis Support System A medical diagnosis system analyzes patient data using the recurrence $T(n) = 2T(n/4) + n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how this affects performance

Code of Conduct

I, MANOJ KUMAR K (192521041) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ASSESSMENT TOOL 1 – STUDENT QUESTION ALLOCATION SHEET

1. Virtual Reality Rendering System

Given recurrence relation:

$$T(n) = 6T(n/3) + n \log n$$

Compare it with the Master Theorem:

$$T(n) = aT(n/b) + f(n)$$

Here,

- $a = 6$
- $b = 3$
- $f(n) = n \log n$

Calculate:

$$n^{(\log_3 6)} \approx n^{1.63}$$

Since $n \log n$ grows slower than $n^{1.63}$,

$$f(n) = O(n^{(1.63 - \varepsilon)}), \text{ where } \varepsilon > 0.$$

Therefore, the recurrence satisfies **Case 1 of the Master Theorem**.

Hence,

$$T(n) = \Theta(n^{(\log_3 6)}) = \Theta(n^{1.63})$$

Rendering Performance Analysis

In the virtual reality rendering system, the scene is divided into **3 smaller sub-scenes**, and **6 recursive rendering operations** are performed. The extra work at each level is $n \log n$, which is smaller than the recursive computation. As the number of objects in the scene increases, the rendering time grows approximately as $n^{1.63}$. This growth is faster than linear but much slower than quadratic, making the algorithm efficient for rendering complex virtual reality scenes.

Conclusion

- **Parameters:** $a = 6$, $b = 3$, $f(n) = n \log n$
- **Master Theorem Case:** Case 1
- **Time Complexity:** $\Theta(n^{1.63})$

Thus, the rendering algorithm has **sub-quadratic asymptotic efficiency** and scales efficiently as scene complexity increases.

2. Medical Diagnosis Support System – Master Theorem

Given recurrence relation:

$$T(n) = 2T(n/4) + n$$

Compare it with the Master Theorem:

$$T(n) = aT(n/b) + f(n)$$

Here,

- $a = 2$
- $b = 4$
- $f(n) = n$

Calculate:

$$n^{(\log_4 2)} = n^{0.5}$$

Now compare $f(n) = n$ with $n^{0.5}$.

Since n grows faster than $n^{0.5}$, and it satisfies the regularity condition:

$$a \times f(n/b) = 2(n/4) = n/2 \leq c \cdot n, \text{ where } c = 1/2 < 1,$$

the recurrence satisfies **Case 3 of the Master Theorem**.

Hence,

$$T(n) = \Theta(n)$$

Performance Analysis

In the medical diagnosis support system, patient data is divided into **2 smaller subproblems**, each of size **$n/4$** . The remaining work at each level is **n** , which dominates the recursive computation. As the number of patient records increases, the execution time grows linearly with the input size. This ensures that the system can process large volumes of medical data efficiently, making it suitable for real-time diagnosis and decision support.

Conclusion

- **Parameters:** $a = 2$, $b = 4$, $f(n) = n$
- **Master Theorem Case:** Case 3
- **Time Complexity:** $\Theta(n)$